

Seagull Lab Controls

A tracker for the pre-start and the race, built on the boats' own telemetry, the course file and the polar. No setting invents a number — every one of them changes what is drawn rather than what is true.

10

RAIL SECTIONS

48

CONTROLS

46

SAVED IN A PRESET

4

PRESETS SHIPPED

At the gun, it changes subject

Before T+0 every overlay asks one question: *where should I be when the gun goes*. After it, none of them mean anything — so the start geometry retires by itself, the camera stops holding the line, and MY BOAT switches from ratio to speed against the polar.

MY BOAT

AUS

RATIO onboard

3.39

+1.59 over target

TO KILL

10.9

s

TO LINE

4.1

s needed

DTL

75

m to line

SOG

62.1

km/h

TWA

-51°

port

BELOW PIN

39

m past the pin

TTK / RATIO AT BOUNDARY in 12.0s · 208 m

-8.4

s to kill

0.26

ratio

LAST TACK 1.20 ratio OUT AND BACK 260 m

6.1

s before you cannot make the line

31.7

s from the pin · +8s tack

FLEET · RATIO NOW

T-15.0

CAN

0.62

ESP

1.29

NZL

1.49

AUS

3.39

MY BOAT, BEFORE THE GUN — Ratio, time to kill, time to line, distance to the line. Then three that look ahead: what your ratio will be when the boundary makes you turn, how many seconds of this course are left before the line becomes unmakeable, and how long the pin-end loop takes. Each is drawn even when it has nothing to say, with the reason in place of the number, so the block never jumps.

PRESETS4>

VIEWfleet>

LOOKlight>

BOATS30s · colours · 15s>

COURSEon>

STARTwave · lines · M1>

ROUTEthis leg · next✓

ROUTEON

NEXT LEGON

WIND FROMthe boat I am tracking▼

DAMPING-10+

LAYLINESon✓

LAYLINESON

WIND FROMthe boat I am tracking▼

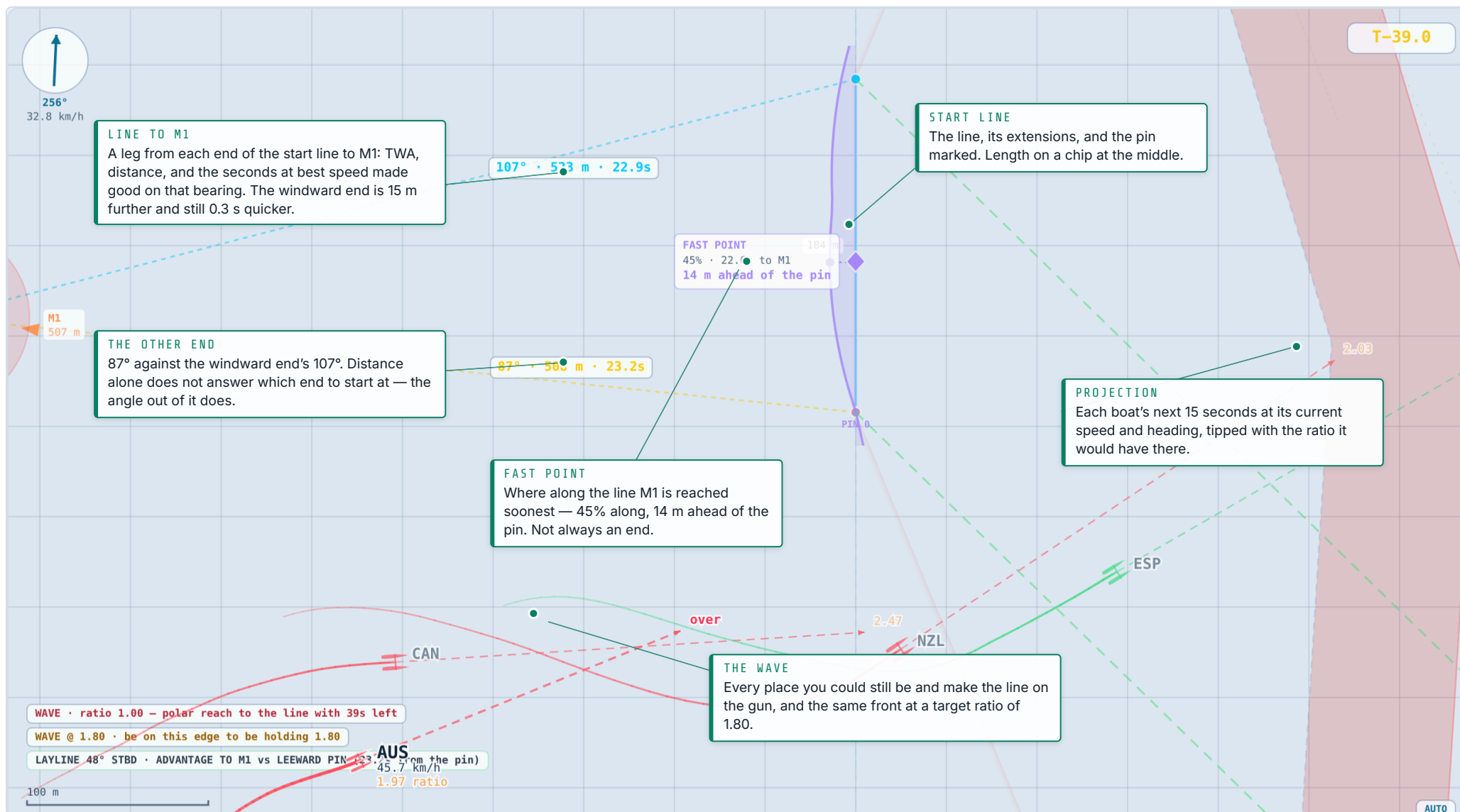
DAMPING-10+

WIDEN-0+

GATE BIASon✓

GATE BIASON

THE RAIL — Every section header carries its own state, so the rail reads shut. A control that depends on another is not drawn at all while that other one is off. Type in the filter box to find a control without knowing which section owns it; HELP puts a one-line explanation back under each row.



T-39 · THE START — Before the gun every overlay asks one question: where should I be at T+0.

NEXT LEG - LEFT MARK
Off the left gate mark on starboard, one gybe, 1779 m. Dashed, so it is never mistaken for the leg you are on.

SECONDS TO THE FIRST TURN
How long that branch runs before the boundary or the layline makes you turn. Always one or the other.

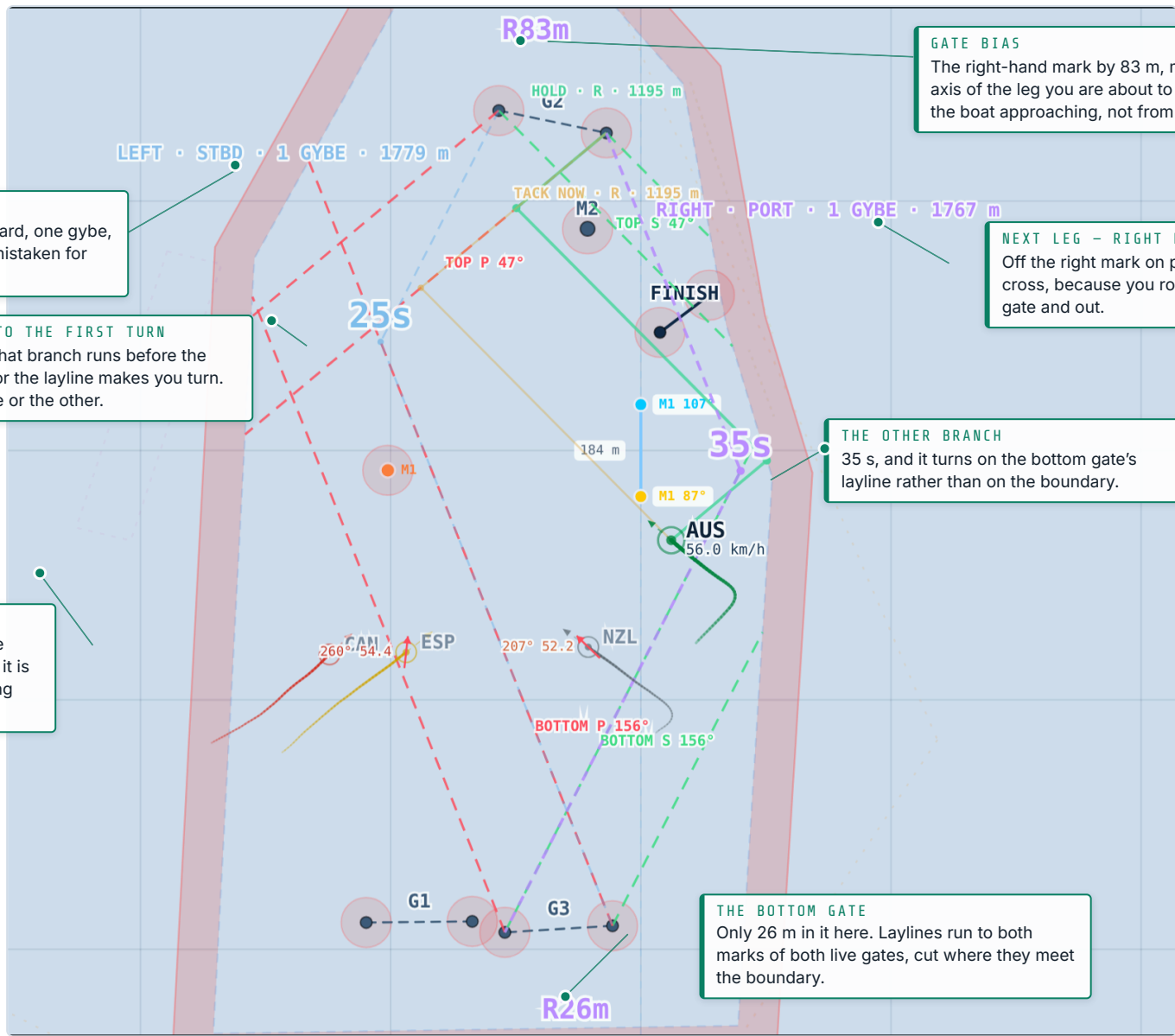
BOUNDARY BAND
The course file stores the inside edge; the 80 m of zone outside it is shaded, because that is the thing you must not enter.

GATE BIAS
The right-hand mark by 83 m, measured along the axis of the leg you are about to sail. Named from the boat approaching, not from the screen.

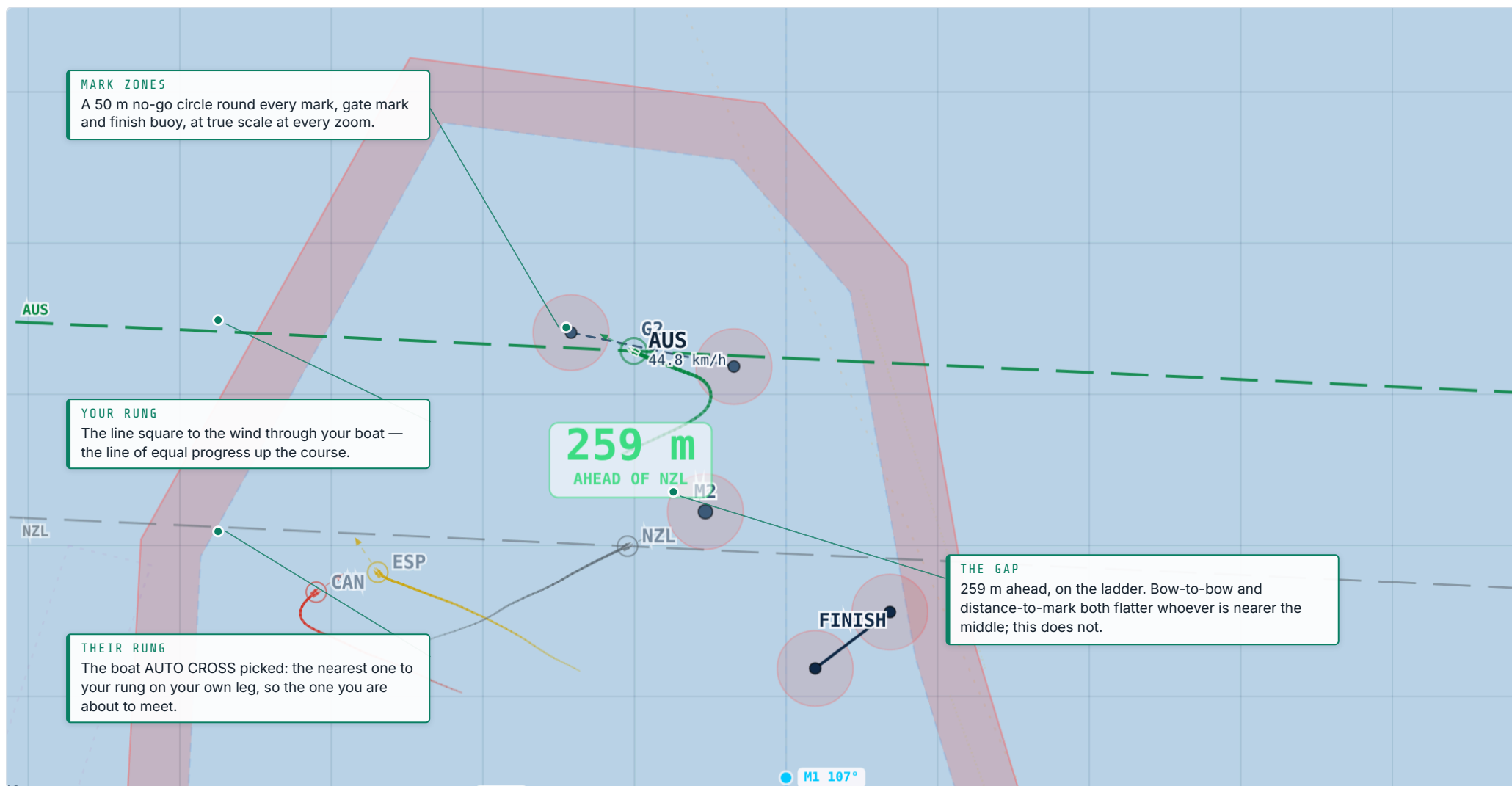
NEXT LEG - RIGHT MARK
Off the right mark on port. The two branches cross, because you round from inside the gate and out.

THE OTHER BRANCH
35 s, and it turns on the bottom gate's layline rather than on the boundary.

THE BOTTOM GATE
Only 26 m in it here. Laylines run to both marks of both live gates, cut where they meet the boundary.



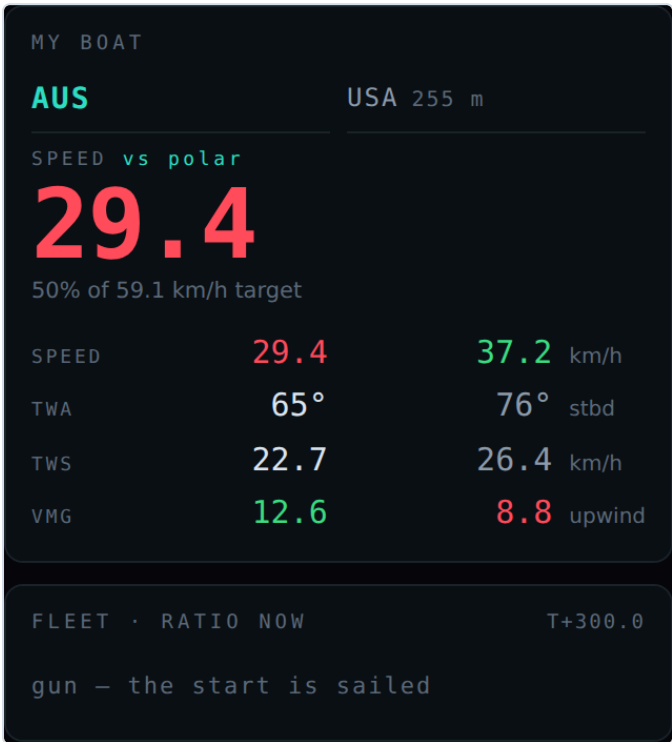
T+149 · ROUTE, NEXT LEG, LAYLINES AND GATE BIAS — Solid green and amber are this leg forked at the boat: hold this gybe, or tack now.



T+239 · CROSSWIND — Who you are actually level with, rather than who looks closer to the mark.

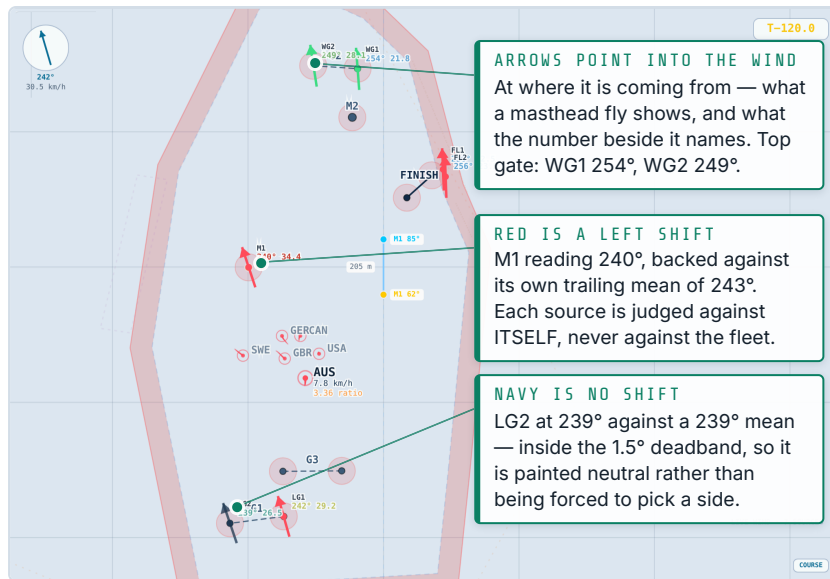
Relatives

Once the gun has gone, MY BOAT stops being about the line. The headline becomes speed against the polar — the one number that says whether the boat is being sailed well, independent of where it is on the course — and beside every row sits the nearest boat you can fairly be compared with.

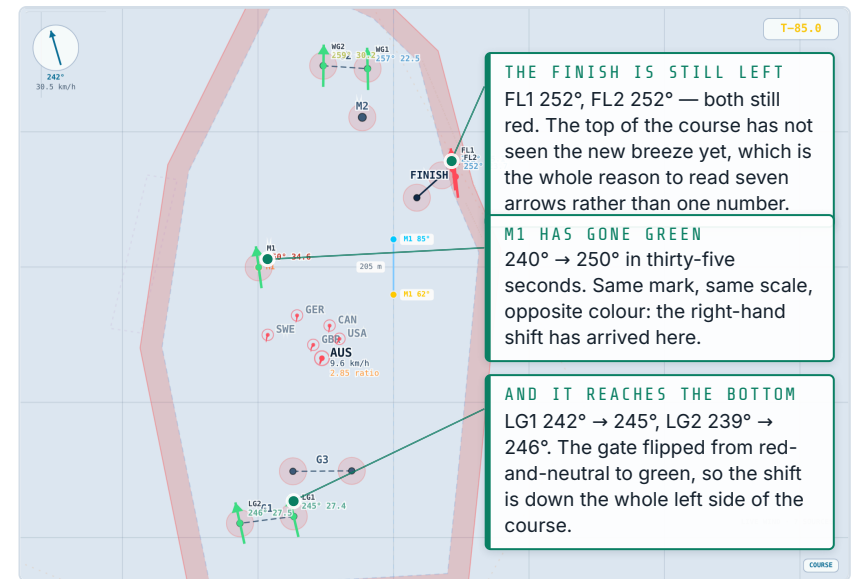


WHO	The nearest boat on your own tack <i>and your own leg</i> , with the distance to it. Tack alone is not a fair comparison: a boat beating at 65° and a boat running at 116° are both on starboard, and setting 29 km/h against 44 would say the running boat is going better when it is simply going downwind. The leg is what makes the row mean something.
SPEED	Yours, theirs, and the percentage of polar target above. Green is the better of the two, red the worse.
TWA	With the tack named under it, so a sign is never left to be interpreted.
TWS	Marked <i>suspect</i> and dimmed where the reading fails the fault gate, rather than quietly averaged in.
VMG	Judged in the sense the leg calls for: upwind, more toward the wind is better; downwind, more away from it.
NOBODY	Where no boat is on your tack and leg, the column reads dashes. No comparison beats a wrong one.





T-120 — Seven mark readings, each with its own colour.



T-85 — Thirty-five seconds later. Same marks, same scale.

RACE 1408 · THE MARKS CHANGE FIRST — a right-hand shift arriving at the bottom of the course while the finish is still in the old breeze. One number for the race would have shown neither.

WHAT THE COLOUR IS MEASURED AGAINST

Each source keeps its own trailing vector mean over $\text{WIND_SHIFT_SEC} = 60 \text{ s}$. The arrow is green when the current reading is right of that mean, red when it is left. Nothing is compared to the fleet, so a mark can be green while the boat beside it is red.

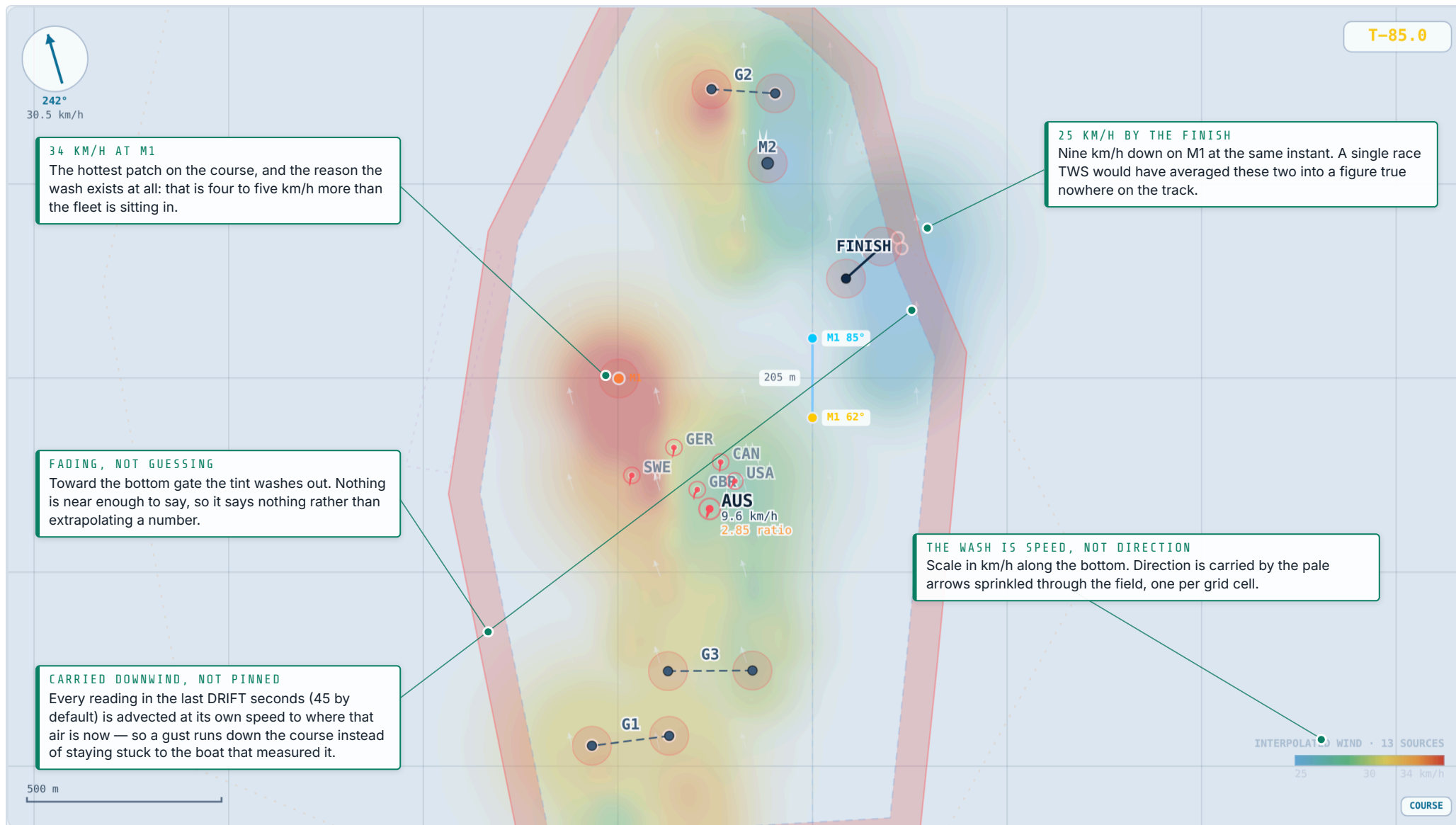
THE DEADBAND

$\text{WIND_SHIFT_DEAD} = 1.5^\circ$. Inside that, the arrow is painted neutral navy instead of being pushed to a side. LG2 at T-120 sits there: 239° against a 239° mean.

WHAT IS DAMPED, AND WHAT IS NOT

These arrows are **raw** — no smoothing at all. The DAMPING sliders in ROUTE, LAYLINES, CROSSWIND and GATE BIAS smooth the wind those overlays consume, not the arrows. The trail behind each source is sampled every 5 s with a 60 s half-life, capped at 220 points.

MARK	T-120	OWN 60 S MEAN	ARROW	T-85	OWN 60 S MEAN	ARROW	Δ TWD
WG1	254°	246°	RIGHT	257°	253°	RIGHT	+3
WG2	249°	247°	RIGHT	259°	252°	RIGHT	+10
FL1	256°	263°	LEFT	252°	257°	LEFT	-4
FL2	256°	264°	LEFT	252°	257°	LEFT	-4
M1	240°	243°	LEFT	250°	243°	RIGHT	+10
LG1	242°	245°	LEFT	245°	243°	RIGHT	+3
LG2	239°	239°	FLAT	246°	241°	RIGHT	+7



T-85 · EVERY SOURCE, INTERPOLATED — WIND FIELD set to *all*: 13 sources — seven marks and six boats — spread across the course, with the LIVE WIND arrows switched off so the field itself is readable. Everything between the readings is inferred, and says so by fading.

Every control

Defaults as shipped. DEP marks a control that only appears while the one above it is on.

VIEW Where the camera is and which way is up		
CONTROL	WHAT IT DOES	DEFAULT
VIEWPORT	Follow fleet · whole course · target boat · target boat offset. The offset puts your boat a third in, with the water it is sailing into ahead of it, and swings round at every rounding.	follow fleet
ORIENTATION	Aligned with course · aligned with wind · target boat heading · north up. Boat-up is averaged over a few seconds so the map does not shake with the yaw.	course
ZOOM	0.1–12. Wheel over the map zooms on the cursor, drag pans, double-click resets.	1
RESET VIEW	Back to the fitted view.	—
SMOOTHING	Off is raw 1 Hz — boats step once a second, where they were measured. 1–2 s runs the map behind the clock so each boat can interpolate.	1 s

LOOK Palette, and the world underneath		
CONTROL	WHAT IT DOES	DEFAULT
THEME	Dark or light, for the whole player — and it carries the water with it.	dark
WATER	Eleven, black through white. The map inverts with it: haloes, grid, hull brightness, captions and chips all follow this one value's luminance.	black
LABELS	Small · medium · large · huge.	medium
FLEET BRIGHTNESS	How far the boats that are not yours sit behind it.	normal
BASEMAP	OpenStreetMap under the course. Needs the network.	off
BASEMAP FADE DEP	How strongly the world shows through.	0.55

BOATS What is drawn on and around each hull		
CONTROL	WHAT IT DOES	DEFAULT
TRAILS	10 · 20 · 30 · 60 s, or everything.	30 s
SPEED LABELS	km/h beside every boat, not only yours.	off
TARGET COLOURS	Above your ratio target red, at or below it green, with a double flash the moment one passes through 1.00.	on
RATIO TARGET DEP	0.2–6. The number the colouring is judged against.	1.80
PROJECTION	Each boat's line run forward at its current speed and heading, with the ratio it would have at the tip.	on
PROJECTION LENGTH DEP	5–60 s.	15 s

COURSE The water you may and may not be in		
CONTROL	WHAT IT DOES	DEFAULT
COURSE LIMITS	The boundary, the marks and their zones.	on
MARK ZONE DEP	0–200 m no-go circle around every mark, gate mark and finish buoy.	50 m
BOUNDARY BAND DEP	0–200 m. The course file stores the inside edge; this shades the zone itself outside it.	80 m

START Everything that retires at T+0		
CONTROL	WHAT IT DOES	DEFAULT
WAVE	Every place you could still be and make the line on the gun.	on
WAVE @ RATIO	The same front at your target ratio, with the band between shaded.	off
WAVE RATIO DEP	0.2–6.	1.80
LINE TO M1	A leg from each end of the line to M1, with its TWA, its length and the seconds it takes at best speed made good on that bearing.	on
M1 WIND FROM DEP	Race TWD · the marks · your boat · boat + marks · all sources. The seconds follow the same wind as the angle.	race TWD
M1 DAMPING DEP	1–120 s.	20 s
LAST TACK RATIO	1–3. The ratio the LAST TACK countdown runs down to. A judgement, not a measurement — hence a control.	1.20
OUT AND BACK TACK	0–30 s added to the pin-end loop for the turn. The polar has nothing to say about a 180°.	8 s
START LAYLINES	The lines to each end of the line, the advantage curve and the fast point.	on
OVERLAYS	Until the gun · always · off. On 'until the gun' the start geometry retires at T+0 and the tool becomes a race tracker.	until the gun

ROUTE Where the corridor takes you		
CONTROL	WHAT IT DOES	DEFAULT
ROUTE	The leg you are on, forked at the boat: hold this gybe, or switch now. Target angles, tacking off the boundary, taking the layline when it comes.	off
NEXT LEG	The leg after, dashed — one branch off each gate mark, on opposite tacks, with the turn count and the seconds to the first turn.	off
WIND FROM DEP	Your masthead · the marks · the race average.	your boat
DAMPING DEP	1–60 s. Short enough to follow a real shift, long enough that the corridor does not shimmer.	10 s

LAYLINES To the two gates you are about to round		
CONTROL	WHAT IT DOES	DEFAULT
LAYLINES	Port and starboard from every mark of the next two gates in course order, at target angle, cut where they meet the boundary. Starboard green, port red.	off
WIND FROM DEP	Your masthead, or the course marks' own readings.	your boat
DAMPING DEP	1–60 s. Undamped, a 1 Hz wind swings a layline several degrees a second.	10 s
WIDEN DEP	–20 to +20°. Opens the laylines out or pulls them in, for when a bad TWS reading hands back the wrong target angle. The route moves with them.	0°

GATE BIAS Which mark is worth metres		
CONTROL	WHAT IT DOES	DEFAULT
GATE BIAS	Metres the favoured mark of each live gate gives you, written outside the gate. L31m means the left-hand mark by 31 m — named from the boat approaching, so screen-left is your right at a leeward gate.	off
WIND FROM DEP	The marks · your boat · boat + marks · all sources.	boat + marks
DAMPING DEP	1–120 s.	20 s

CROSSWIND Who you are actually level with		
CONTROL	WHAT IT DOES	DEFAULT
CROSSWIND	The ladder rung through your boat — the line square to the wind. Anyone on it is level with you, however far across the course they are.	off
AUTO CROSS DEP	The nearest boat to your rung on your own leg: the gap in metres, green ahead, red behind, plus the seconds to a crossing on opposite tacks.	on
WIND FROM DEP	Race TWD · your boat · the marks. The race TWD is the default because a rung is read a metre at a time and a live masthead swings it several degrees a second.	race TWD
DAMPING DEP	1–120 s.	30 s

WIND Readings, and the field between them		
CONTROL	WHAT IT DOES	DEFAULT
LIVE WIND	Off · marks · boats · all sources. Raw readings as arrows pointing into the wind, red for a left shift against their own last minute and green for a right one.	off
WIND FIELD	Off · marks · boats · all sources. Those readings interpolated across the course, fading where nothing is near enough to say.	off
DRIFT DEP	0–120 s of past readings carried downwind to where that air is now, so a gust travels down the course instead of staying stuck to the boat that measured it.	45 s

Seagull Lab · a static site, no build step. The rail is one declared array — a new control arrives with its label, its widget and its wiring, and joins the presets automatically.